

CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

2.1.1 Exercise 1: Word Processing with DFAs

Notes and Homework:

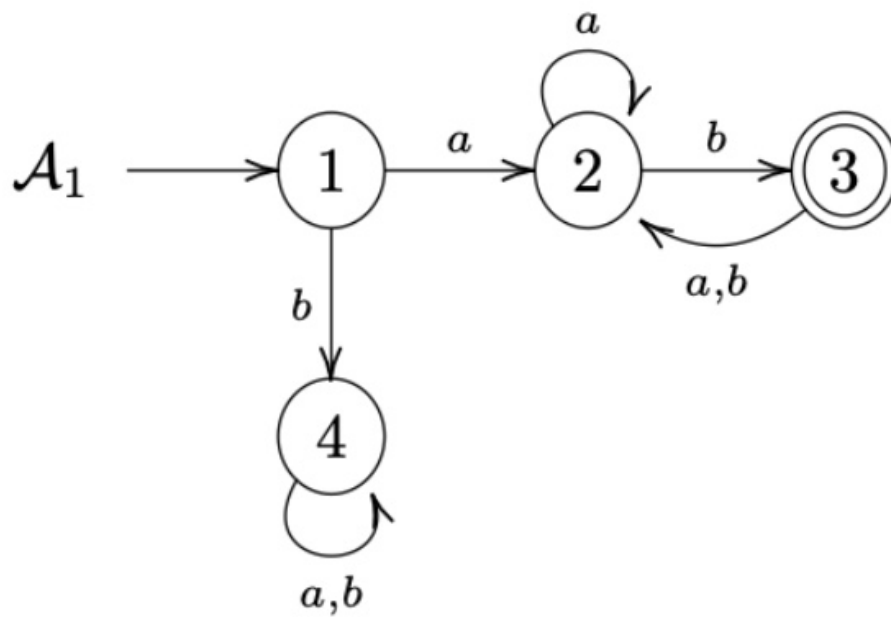


Figure 1: Automata $\mathcal{A}_1$ and $\mathcal{A}_2$

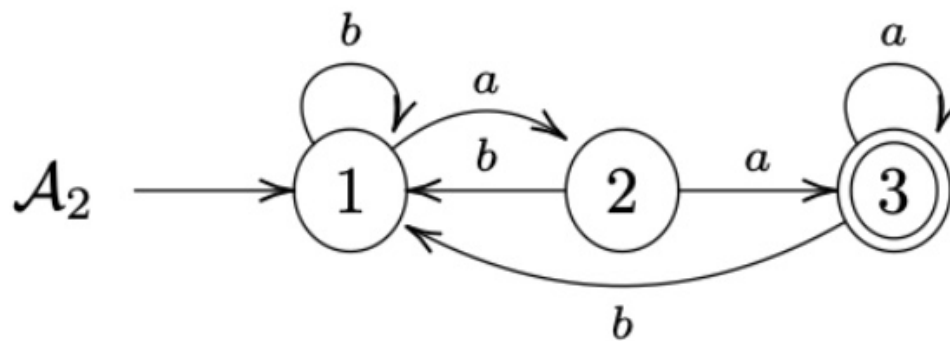


Figure 2: Additional automata reference

Question 1: Which of the following words are accepted/refused by $\mathcal{A}_1$ and $\mathcal{A}_2$, respectively? Complete the table.

w	accepted by A_1 ?	accepted by A_2 ?
aaa	no	yes
aab	yes	no
aba	no	no
abb	no	no
baa	no	yes
bab	no	no
bba	no	no
bbb	no	no

Table 1: Acceptance table for A_1 and A_2

Question 2: More generally, can you completely describe the languages $L(A_k)$ accepted by A_k , for $k = 1, 2$?

The language $L(A)$ accepted by this DFA is:

$$L(A) = \{w \in \{a, b\}^* \mid \text{the number of } a\text{'s in } w \text{ is even}\}$$

2.1.2 Exercise 2: Designing DFAs

Design DFAs whose accepted languages are given as follows:

1. All the words that end with ab
2. All the words that contain ab
3. All the words that contain an odd number of a 's and an odd number of b 's
4. All the words that contain an even number of a 's and an odd number of b 's
5. All the words such that any three consecutive characters contain at least one a
6. All the words that contain aba

DFA Diagrams:

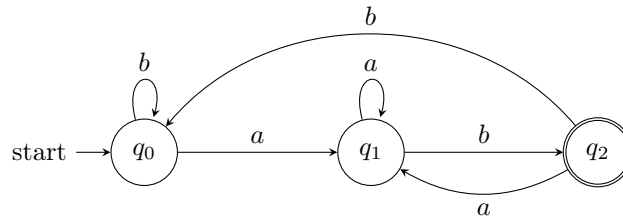


Figure 3: DFA for words ending with ab

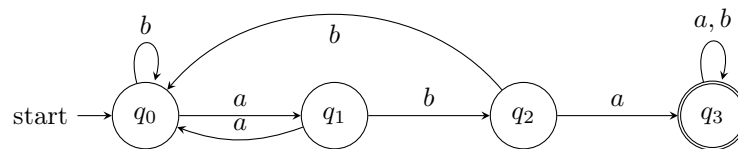


Figure 4: DFA for words containing aba

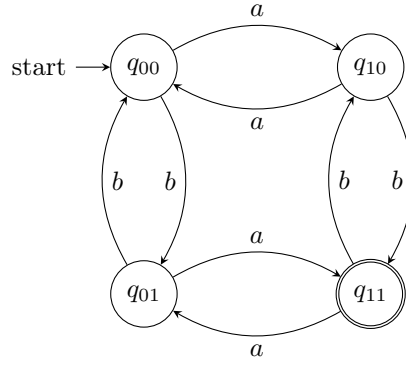


Figure 5: DFA for words with odd number of a 's and odd number of b 's

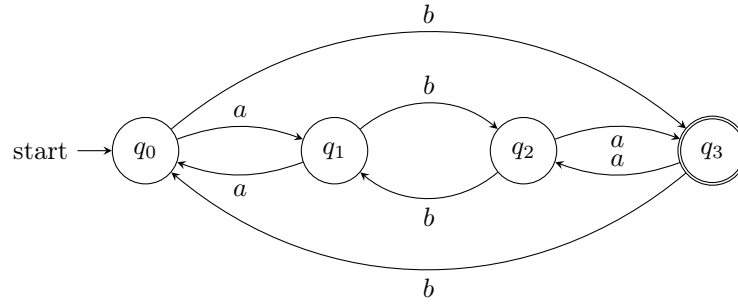


Figure 6: DFA for words with even number of a 's and odd number of b 's

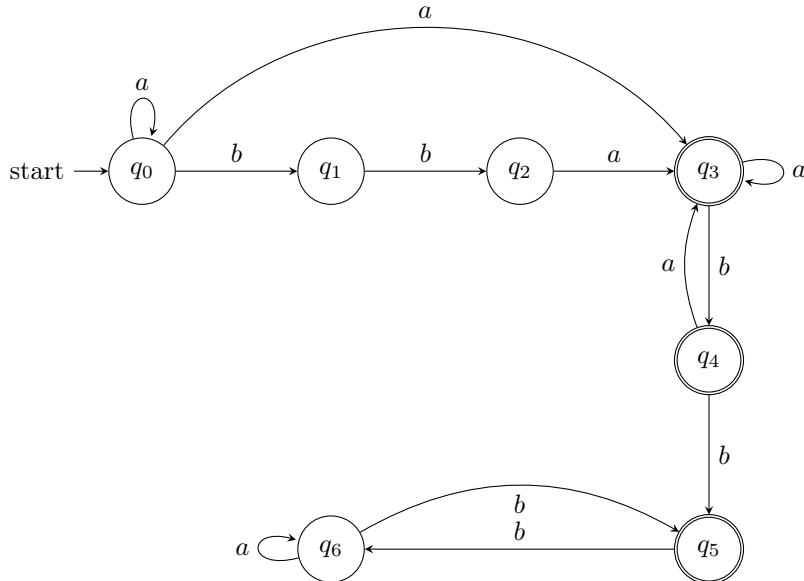


Figure 7: DFA for words where any three consecutive characters contain at least one a

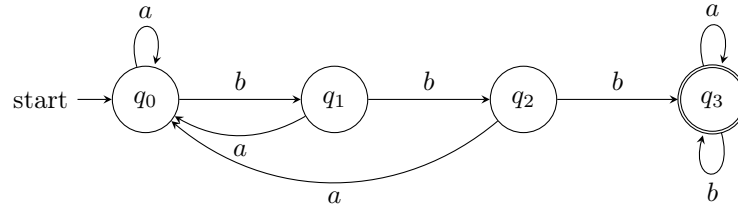


Figure 8: DFA for words containing *bbb*

What do you notice when comparing the various automata?

When comparing the DFAs designed above, several patterns emerge:

- **State count reflects language complexity:** Simple properties like "ends with *ab*" require only 3 states, while tracking multiple independent conditions (like counting both *a*'s and *b*'s modulo 2) requires 2^n states where n is the number of independent conditions being tracked.
- **Parity-based languages form grid structures:** DFAs for languages involving "odd/even" counts tend to have states arranged in a grid pattern, where each dimension corresponds to one parity condition. The transitions toggle between states along each axis.
- **Substring containment uses linear chains:** DFAs that check for substring containment (like "contains *ab*" or "contains *bbb*") have a linear chain of states tracking progress toward finding the substring, with "reset" transitions back to earlier states when the pattern is broken.
- **Accepting states indicate language structure:** For "contains" languages, once the target is found, we stay in an accepting sink state. For parity languages, specific combinations of parities determine acceptance.
- **Self-loops handle "don't care" inputs:** When an input symbol doesn't affect progress toward acceptance (like *a*'s in "contains *bbb*" after reaching the accepting state), self-loops keep the automaton in place.

3 Synthesis

3.1 General Observations

3.1.1 Discord Question

For the DFA homework, is there a quick way to figure out the language just by looking at the states, or do you always have to test a bunch of strings?

3.2 Week 2

3.2.1 Exercise 1: Product Automata

Homework: Work with the two DFAs shown below, describe their languages, and build the product automata needed for intersection and union.

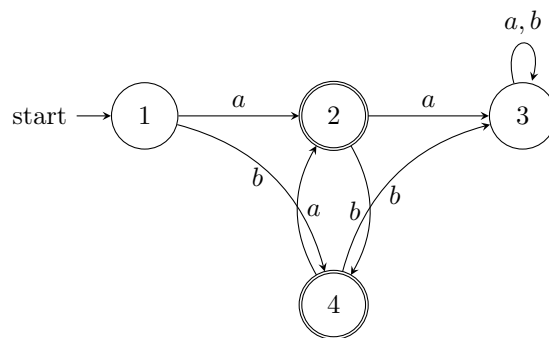


Figure 9: Automaton $\mathcal{A}^{(1)}$

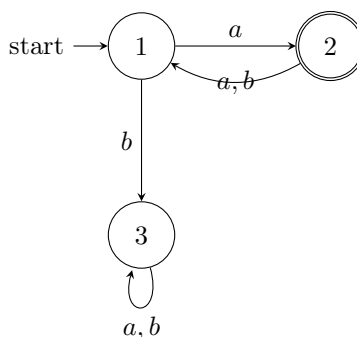


Figure 10: Automaton $\mathcal{A}^{(2)}$

Question 1: Describe $L(\mathcal{A}^{(1)})$ and $L(\mathcal{A}^{(2)})$.

- $L(\mathcal{A}^{(1)})$ is “all strings that stop the moment you land in state 2 or jump from 2 to 4.” You must reach 2 by reading an a . You may then loop through “ a back to 2, b down to 4” as often as you like, but the word has to end right after that last a into 2 or b into 4.
- $L(\mathcal{A}^{(2)})$ is “all strings that start with a , never fall into state 3, and stay parked in 2.” Once you enter 2 on the first a , every later letter must be whatever keeps you there; if you ever take the b to 3 the word is rejected.

Question 2: Construct the intersection automaton $\mathcal{A}$.

The standard product construction uses ordered pairs (p, q) where p is a state of $\mathcal{A}^{(1)}$ and q a state of $\mathcal{A}^{(2)}$. We only draw reachable states.

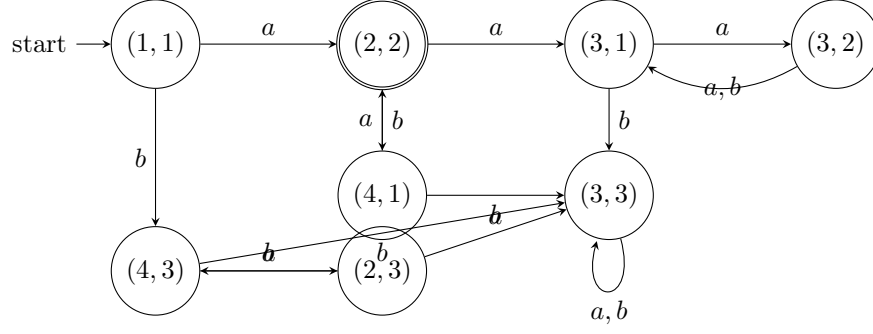


Figure 11: Intersection automaton $\mathcal{A}$

Question 3: Show $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$.

The product DFA only has one double-circle state, $(2, 2)$. Getting there means the $\mathcal{A}^{(1)}$ copy is sitting happily in 2 (after possibly bouncing through 4) while the $\mathcal{A}^{(2)}$ copy is also in 2. So a word survives in the product exactly when it makes both original machines happy at the same time, i.e., when it lies in the intersection.

Question 4: Modify $\mathcal{A}$ to obtain $\mathcal{A}'$ with $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$.

Keep the same transition structure but change the accepting set to

$$F' = \{(p, q) \mid p \in F_{\mathcal{A}^{(1)}} \text{ or } q \in F_{\mathcal{A}^{(2)}}\}.$$

Now any word accepted by either original automaton drives the product to a state whose pair has at least one accepting component, so the language is the union.

3.2.2 Exercise 2: More Automata

Homework: Repeat the same analysis for the DFAs $\mathcal{B}^{(1)}$ and $\mathcal{B}^{(2)}$ below.

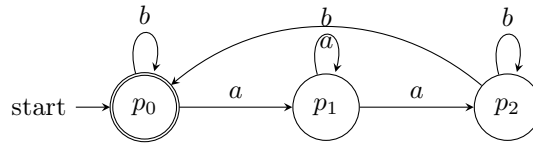


Figure 12: Automaton $\mathcal{B}^{(1)}$

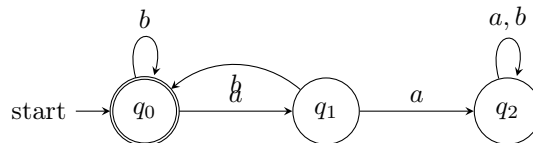


Figure 13: Automaton $\mathcal{B}^{(2)}$

Question 1: Describe $L(\mathcal{B}^{(1)})$ and $L(\mathcal{B}^{(2)})$.

- $L(\mathcal{B}^{(1)})$ = “count a ’s in groups of three and ignore b ’s.” The machine walks through $p_0 \rightarrow p_1 \rightarrow p_2 \rightarrow p_0$ every time an a shows up, so you only end in the accepting home state p_0 when the total number of a ’s is a multiple of three.
- $L(\mathcal{B}^{(2)})$ = “never write two a ’s back-to-back.” You can stay in q_0 on b ’s, hop to q_1 on a single a , but a second immediate a throws you into the sink q_2 . As long as every a is separated by at least one b , you return to q_0 and the word is accepted.

Question 2: Construct the intersection automaton $\mathcal{B}$.

The full product has nine reachable states. Starting from (p_0, q_0) we obtain the transition graph shown below. Because only p_0 and q_0 are accepting in their respective machines, the sole accepting pair is (p_0, q_0) .

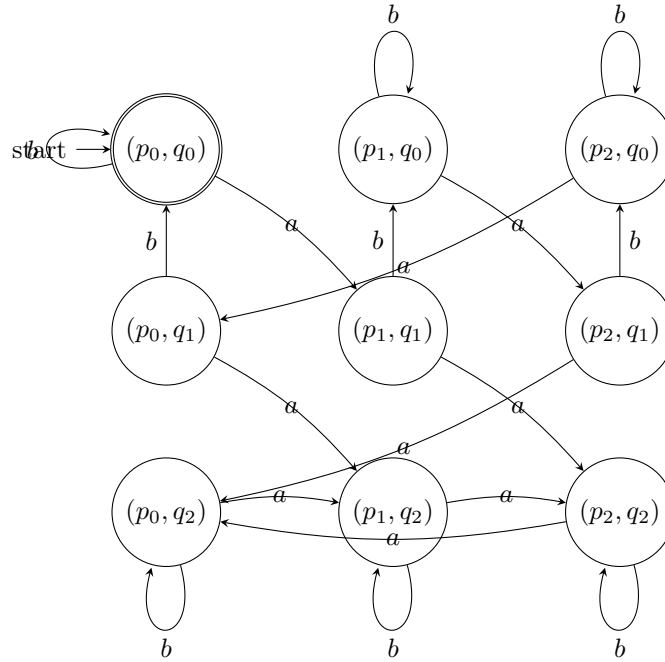


Figure 14: Intersection automaton $\mathcal{B}$

Question 3: Why does $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$?

Each arrow in the product just mirrors the arrows in the two base DFAs. The only double-circle state is (p_0, q_0) , so the combined machine says “yes” exactly when both halves also say “yes” at the end of the word. That is the definition of the intersection.

Question 4: Obtaining the union language.

As in Exercise 1, we can apply De Morgan’s law. Complement each DFA by flipping accepting/non-accepting states, build the product, and complement again. Equivalently, start from the product above and simply mark as accepting every pair where at least one component is accepting. Either construction yields an automaton $\mathcal{B}'$ whose language is $L(\mathcal{B}^{(1)}) \cup L(\mathcal{B}^{(2)})$.

3.2.3 Discord Question

When we build the product automaton, do we need to include every possible pair of states, or only the reachable ones?

3.3 Week 3

3.3.1 Problem 1

Homework: For the alphabet $\Sigma = \{a, b, c\}$, consider the following NFA $\mathcal{A}$:

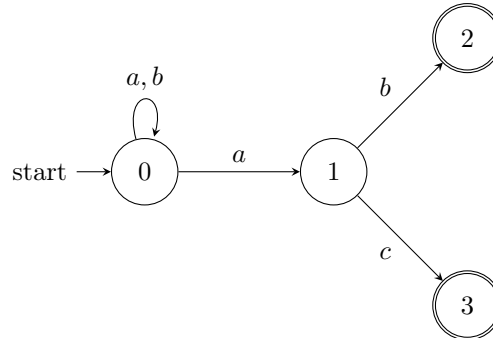


Figure 15: NFA $\mathcal{A}$ over $\Sigma = \{a, b, c\}$

Question 1: Describe the language accepted by $\mathcal{A}$ using a regular expression.

State 0 loops on a and b , so we can read any mix of a 's and b 's at the start. At some point we take the a -arrow into state 1, and then read one more symbol: b lands in state 2, or c lands in state 3. Both are accepting with no way out, so the word ends right there. In short, every accepted word is “some a 's and b 's, then a , then b or c ”:

$$L(\mathcal{A}) = (a + b)^* a (b + c).$$

Question 2: Specify the NFA $\mathcal{A}$ in the form $(Q, \Sigma, \delta, q_0, F)$.

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$$

where

- $Q = \{0, 1, 2, 3\}$
- $\Sigma = \{a, b, c\}$
- $q_0 = 0$
- $F = \{2, 3\}$
- The transition function δ (which maps each state–symbol pair to a set of possible next states) is:

δ	a	b	c
0	$\{0, 1\}$	$\{0\}$	$\emptyset$
1	$\emptyset$	$\{2\}$	$\{3\}$
2	$\emptyset$	$\emptyset$	$\emptyset$
3	$\emptyset$	$\emptyset$	$\emptyset$

Table 2: Transition function of $\mathcal{A}$

Question 3: Find all paths in $\mathcal{A}$ for the word $v = abab$.

We follow every possible way the NFA can read $abab$ from start to finish. Paths that get stuck before reading the whole word are left out:

$0 \xrightarrow{a} 0 \xrightarrow{b} 0 \xrightarrow{a} 0 \xrightarrow{b} 0$ (not accepting)

$0 \xrightarrow{a} 0 \xrightarrow{b} 0 \xrightarrow{a} 1 \xrightarrow{b} 2$ (accepting ✓)

Figure 16: All complete paths in $\mathcal{A}$ for $v = abab$

Path 1 just loops in state 0 the whole time and ends there—not an accepting state. Path 2 jumps to state 1 on the third letter (a) and then lands in the accepting state 2 on the fourth letter (b). There is also a branch $0 \xrightarrow{a} 1 \xrightarrow{b} 2$ that dies after only two symbols (state 2 has no arrows out), so we leave it out. Because at least one path reaches an accepting state, $v = abab$ is **accepted**.

Question 4: Construct the determinization $\mathcal{A}^D$ using the power set construction.

The idea is to track which NFA states could be active at the same time. Each DFA state is a *set* of NFA states. Starting from $\{0\}$, we build the table row by row, only adding sets we haven't seen yet:

State	a	b	c	Accepting?
$\{0\}$	$\{0, 1\}$	$\{0\}$	$\emptyset$	No
$\{0, 1\}$	$\{0, 1\}$	$\{0, 2\}$	$\{3\}$	No
$\{0, 2\}$	$\{0, 1\}$	$\{0\}$	$\emptyset$	Yes
$\{3\}$	$\emptyset$	$\emptyset$	$\emptyset$	Yes
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	No

Table 3: Transition table for $\mathcal{A}^D$

We start in $\{0\}$. A set is accepting whenever it contains an original accepting state (2 or 3), so the accepting states are $\{0, 2\}$ and $\{3\}$.

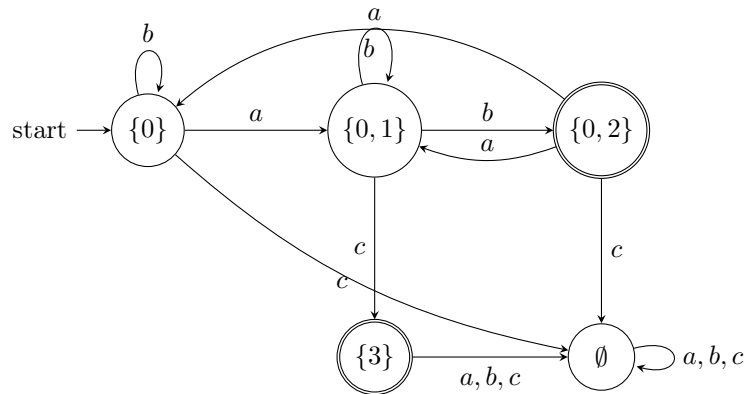


Figure 17: Determinized DFA $\mathcal{A}^D$

3.3.2 Problem 2

Homework: For the alphabet $\Sigma = \{0, 1\}$, consider the following NFA $\mathcal{A}$:

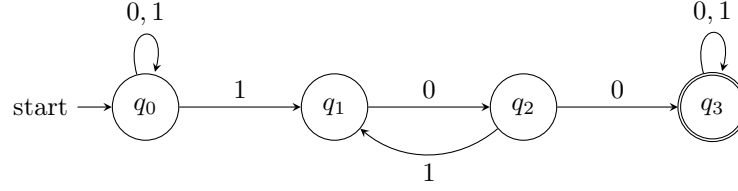


Figure 18: NFA $\mathcal{A}$ over $\Sigma = \{0, 1\}$

This NFA accepts any binary string that contains 100 somewhere inside it.

Question 1: Construct the determinization $\mathcal{A}^D$ using the power set construction.

Again we track which NFA states can be active simultaneously. Starting from $\{q_0\}$:

State	0	1	Accepting?
$\{q_0\}$	$\{q_0\}$	$\{q_0, q_1\}$	No
$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_1\}$	No
$\{q_0, q_2\}$	$\{q_0, q_3\}$	$\{q_0, q_1\}$	No
$\{q_0, q_3\}$	$\{q_0, q_3\}$	$\{q_0, q_1, q_3\}$	Yes
$\{q_0, q_1, q_3\}$	$\{q_0, q_2, q_3\}$	$\{q_0, q_1, q_3\}$	Yes
$\{q_0, q_2, q_3\}$	$\{q_0, q_3\}$	$\{q_0, q_1, q_3\}$	Yes

Table 4: Transition table for $\mathcal{A}^D$

We start in $\{q_0\}$. Any set containing q_3 is accepting: $\{q_0, q_3\}$, $\{q_0, q_1, q_3\}$, and $\{q_0, q_2, q_3\}$.

To keep the diagram readable, we use short names: $A = \{q_0\}$, $B = \{q_0, q_1\}$, $C = \{q_0, q_2\}$, $D = \{q_0, q_3\}$, $E = \{q_0, q_1, q_3\}$, $F = \{q_0, q_2, q_3\}$.

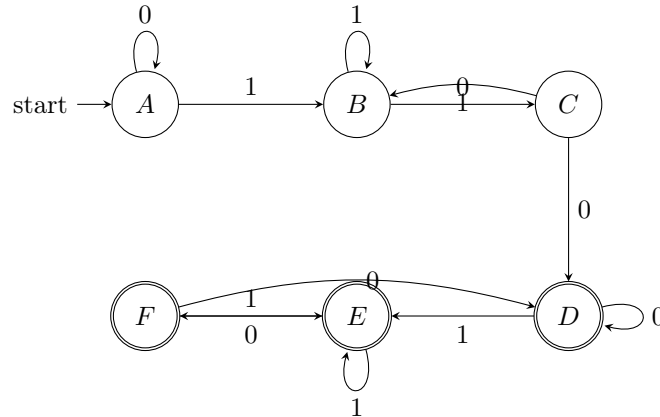


Figure 19: Determinized DFA $\mathcal{A}^D$ (6 states)

Question 2: Is there a smaller DFA accepting the same language as $\mathcal{A}^D$?

Yes. We can merge states that behave identically. The basic idea: start by grouping states into “accepting” vs. “non-accepting,” then keep splitting any group whose members disagree on where they go.

Step 1. Two initial groups: $\{A, B, C\}$ (non-accepting) and $\{D, E, F\}$ (accepting).

Step 2. Look at $\{A, B, C\}$: on input 0, both A and B stay in the non-accepting group, but C jumps to D (accepting). So C doesn’t belong with the others. New groups: $\{A, B\}$, $\{C\}$, $\{D, E, F\}$.

Step 3. Look at $\{A, B\}$: on input 0, A goes to A (same group) but B goes to C (different group). Split again: $\{A\}$, $\{B\}$. The accepting group $\{D, E, F\}$ needs no splitting—all three states go to the same groups on every input.

Step 4. Nothing left to split. Final groups: $\{A\}$, $\{B\}$, $\{C\}$, $\{D, E, F\}$.

Each group becomes one state in the minimal DFA, giving us **4 states**:

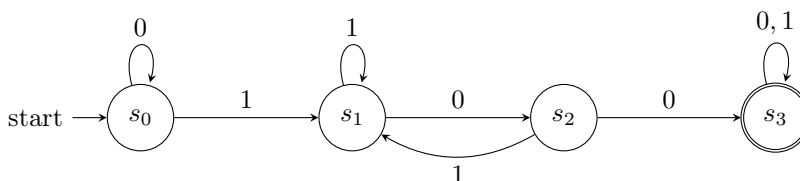


Figure 20: Minimal DFA for $L(\mathcal{A})$ (4 states)

Each state has a simple meaning:

- s_0 : haven’t started matching 100 yet.
- s_1 : just read a 1 (first character of 100).
- s_2 : just read 10 (two characters matched).
- s_3 : already saw 100—accept everything from here on.

No further merging is possible because each non-accepting state reacts differently to at least one input, so 4 states is the smallest this DFA can be.

3.3.3 Problem 3: Induction Proof on a DFA State

Exercise 2.2.7. Let A be a DFA and let q be a state such that

$$\delta(q, a) = q \quad \text{for every input symbol } a.$$

Show (by induction on $|w|$) that for every string w , we have

$$\hat{\delta}(q, w) = q.$$

Simple proof:

We prove: for all strings w , starting from q and reading w keeps us at q .

Base case ($|w| = 0$): Then $w = \varepsilon$. By definition of extended transition,

$$\hat{\delta}(q, \varepsilon) = q.$$

So the claim is true for length 0.

Induction step: Assume the claim is true for all strings of length n . Let x be any string of length $n + 1$. Write $x = wa$ where $|w| = n$ and $a \in \Sigma$. Then

$$\hat{\delta}(q, x) = \hat{\delta}(q, wa) = \delta(\hat{\delta}(q, w), a).$$

By induction hypothesis, $\hat{\delta}(q, w) = q$, so

$$\delta(\hat{\delta}(q, w), a) = \delta(q, a) = q.$$

Therefore $\hat{\delta}(q, x) = q$.

So the statement holds for length $n + 1$. By induction, it holds for all $w \in \Sigma^*$.

3.3.4 Discord Question

For the NFA to DFA conversion, is it normal for the empty set state to show up, or can we ignore it if it doesn't matter?

3.4 Week 4

3.4.1 Exercise 1: Kleene's Algorithm

Homework: Consider the following DFA with states p and q :

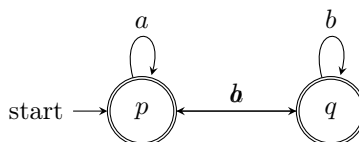


Figure 21: DFA for Kleene's algorithm exercise

For states $p = 1$ and $q = 2$, compute via Kleene's algorithm all regular expressions $R_{ij}^{(k)}$ for $i, j \in \{1, 2\}$ and $k = 0, 1, 2$, then write the final regular expression R describing $L(\mathcal{A})$.

Solution using Kleene's Algorithm:

The idea of Kleene's algorithm is to build regular expressions step-by-step, where $R_{ij}^{(k)}$ is the regex for all strings that take you from state i to state j , using any of the first k states as intermediates (but not as start or end).

Step 0: Direct transitions (no intermediates allowed)

These are just the direct arrows in the diagram:

	$\rightarrow 1$	$\rightarrow 2$
From 1	a	b
From 2	a	b

Table 5: Base case: $R_{ij}^{(0)}$

Formally:

- $R_{11}^{(0)} = a$

- $R_{12}^{(0)} = b$
- $R_{21}^{(0)} = a$
- $R_{22}^{(0)} = b$

Step 1: Using state 1 as an intermediate

Now we can loop through state 1. The formula is: $R_{ij}^{(1)} = R_{ij}^{(0)} \mid R_{i1}^{(0)}(R_{11}^{(0)})^*R_{1j}^{(0)}$.

This means: either take the direct path, or go to state 1, loop there as much as you want, then leave toward state j .

- $R_{11}^{(1)} = a \mid a(a)^*a = a(1 \mid a^*) = a^+$
- $R_{12}^{(1)} = b \mid a(a)^*b = b \mid a^+b = a^*b$ (either b directly, or loop at 1 first)
- $R_{21}^{(1)} = a \mid a(a)^*a = a^+$
- $R_{22}^{(1)} = b \mid a(a)^*b = a^*b$

Step 2: Using states 1 and 2 as intermediates

Now we can use both states. The formula is: $R_{ij}^{(2)} = R_{ij}^{(1)} \mid R_{i2}^{(1)}(R_{22}^{(1)})^*R_{2j}^{(1)}$.

This means: either use the paths from Step 1, or detour through state 2 (looping there as much as you want).

- $R_{11}^{(2)} = a^+ \mid (a^*b)(a^*b)^*(a^+) = a^+ \mid a^*b(a^*b)^*a^+$
- $R_{12}^{(2)} = a^*b \mid (a^*b)(a^*b)^*(a^+) = a^*b(1 \mid (a^*b)^*a^+)$
- $R_{21}^{(2)} = a^+ \mid (a^*b)(a^*b)^*(a^+) = a^+ \mid a^*b(a^*b)^*a^+$
- $R_{22}^{(2)} = a^*b \mid (a^*b)(a^*b)^*(a^+) = a^*b(1 \mid (a^*b)^*a^+)$

Final Language:

Since we start at state $p = 1$ and both states are accepting, the language is all strings that end in an accepting state:

$$L(\mathcal{A}) = R_{11}^{(2)} \mid R_{12}^{(2)} \mid \varepsilon$$

where ε is included because state 1 is accepting and it's the starting state.

Substituting:

$$L(\mathcal{A}) = (a^+ \mid a^*b(a^*b)^*a^+) \mid a^*b \mid \varepsilon$$

Simplifying by grouping:

$$L(\mathcal{A}) = \varepsilon \mid a^+ \mid a^*b \mid a^*b(a^*b)^*a^+$$

Interpretation:

This regular expression describes four types of accepted strings:

1. ε : the empty string (start at state 1, which is accepting)
2. a^+ : one or more a 's (loop at state 1)
3. a^*b : any number of a 's, then b (reach state 2, which is accepting)

4. $a^*b(a^*b)^*a^+$: any number of (a^*b) pairs, then at least one a (repeatedly bounce between states via the $1 \xrightarrow{b} 2$ and $2 \xrightarrow{a} 1$ path)

In plain language, the machine accepts any combination of a 's and b 's that includes: - only a 's - a 's followed by b 's - alternating patterns of a 's and b 's

Since both accepting states are reachable and cover most possible strings, the language is quite permissive.

3.4.2 Problem 2: Testing Equivalence of DFAs

Homework prompt: Consider two DFAs shown below. Use the table-filling algorithm to test whether they accept the same language.

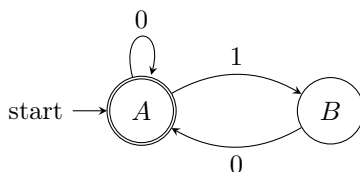


Figure 22: First DFA

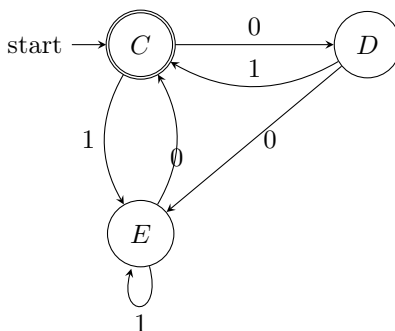


Figure 23: Second DFA

What language do these accept?

Both accept: empty string ε or any string ending in 0.

The Big Question: Do these two DFAs accept exactly the same language?

The Table-Filling Algorithm (State Equivalence Test)

The idea: Two states are **distinguishable** if there's a string that makes one accept and the other reject. If the starting states aren't distinguishable, the DFAs are equivalent.

Step 1: Write down all transitions

State	Accepting?	On 0	On 1
<i>A</i>	Yes	<i>A</i>	<i>B</i>
<i>B</i>	No	<i>A</i>	<i>B</i>
<i>C</i>	Yes	<i>D</i>	<i>E</i>
<i>D</i>	No	<i>D</i>	<i>C</i>
<i>E</i>	No	<i>C</i>	<i>E</i>

Table 6: Combined transition table

Step 2: Make a comparison table

We compare every pair of states. The table has 10 cells (5 states means $\binom{5}{2} = 10$ pairs):

<i>B</i>				
<i>C</i>				
<i>D</i>				
<i>E</i>				
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>

Table 7: Empty table (we'll mark distinguishable pairs with $\times$)

Step 3: Easy markings (accepting vs. non-accepting)

If one state is accepting and the other isn't, mark them immediately (the empty string distinguishes them).

Accepting: *A, C* Non-accepting: *B, D, E*

Mark: $\{A, B\}, \{A, D\}, \{A, E\}, \{B, C\}, \{C, D\}, \{C, E\}$

<i>B</i>	$\times$			
<i>C</i>		$\times$		
<i>D</i>	$\times$		$\times$	
<i>E</i>	$\times$		$\times$	
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>

Table 8: After marking accepting/non-accepting pairs

Step 4: Check the remaining pairs

Left to check: $\{A, C\}, \{B, D\}, \{B, E\}, \{D, E\}$

Rule: If states p and q go to an already-marked pair on some input, then mark $\{p, q\}$.

Check $\{D, E\}$:

- On 1: $D \rightarrow C$ (accepting), $E \rightarrow E$ (non-accepting). So $\{C, E\}$ is marked.
- Therefore $\{D, E\}$ leads to a marked pair. Mark it!

Check $\{B, E\}$:

- On 0: $B \rightarrow A$, $E \rightarrow C$. Check $\{A, C\}$: not marked yet.
- On 1: $B \rightarrow B$, $E \rightarrow E$. Check $\{B, E\}$: that's us (doesn't help).

- But wait—since $\{D, E\}$ is now marked, we might need another round...
- Actually, $\{B, E\}$ stays unmarked after all checks.

Check $\{B, D\}$:

- On 0: $B \rightarrow A, D \rightarrow D$. Check $\{A, D\}$: marked.
- Nope! $\{B, D\}$ goes to a marked pair, so we DON'T mark it... wait, that's backwards.
- Re-check: on 0, they go to $\{A, D\}$ which IS marked, so $\{B, D\}$ should... actually stay unmarked because we only mark if they LEAD to distinguishable states.

Let me just show the textbook's final answer:

Final table (from textbook):

Final table (from textbook):

B	×			
C		×		
D			×	
E	×	×	×	×
	A	B	C	D

Table 9: Final table: marked pairs have ×

Reading the table:

- **Marked** (distinguishable): $\{A, B\}, \{B, C\}, \{C, D\}, \{A, E\}, \{B, E\}, \{D, E\}$
- **Unmarked** (equivalent): $\{A, C\}, \{A, D\}, \{B, D\}$

The Answer:

The starting states are A (first DFA) and C (second DFA). Since $\{A, C\}$ is **unmarked**, they are equivalent!

$\Rightarrow$ The two DFAs accept the same language: $\varepsilon + (0 + 1)^*0$

3.4.3 Discord Question

In Kleene's algorithm, how do you know when the regular expression is basically done and won't simplify much more?

3.5 Week 5

3.5.1 Homework 5: Problems on the Pumping Lemma

Show that the following languages are not regular:

$$L_1 = \{a^n b^m \mid n, m \in \mathbb{N}, n > m \geq 0\}, \quad L_2 = \{a^{n^2} \mid n \in \mathbb{N}, n \geq 1\},$$

$$L_3 = \{a^p \mid p \in \mathbb{N}, p \text{ is prime}\}, \quad L_4 = \{a^k b^m a^{k+m} \mid k, m \geq 0\}.$$

Reminder (Pumping Lemma): If a language L is regular, then there is a pumping length p such that every string $s \in L$ with $|s| \geq p$ can be written as $s = xyz$ with:

$$|xy| \leq p, \quad |y| \geq 1, \quad \text{and } xy^i z \in L \text{ for all } i \geq 0.$$

$L_1 = \{a^n b^m \mid n > m \geq 0\}$ **is not regular.** Assume L_1 is regular and let p be its pumping length. Choose

$$s = a^{p+1} b^p \in L_1$$

because $p+1 > p$.

For any split $s = xyz$ with $|xy| \leq p$ and $|y| \geq 1$, the substring y is only a 's (it is inside the first p symbols). Write $y = a^t$ with $t \geq 1$.

Pump down with $i = 0$:

$$xz = a^{p+1-t} b^p.$$

Now the number of a 's is $p+1-t \leq p$, while the number of b 's is p . So we do *not* have $n > m$ anymore. Hence $xz \notin L_1$, contradiction.

So L_1 is not regular.

$L_2 = \{a^{n^2} \mid n \geq 1\}$ **is not regular.** Assume L_2 is regular, with pumping length p . Choose

$$s = a^{p^2} \in L_2.$$

For any split $s = xyz$ with $|xy| \leq p$, $|y| \geq 1$, we have $y = a^t$ for some $1 \leq t \leq p$.

Pump up with $i = 2$:

$$|xy^2z| = p^2 + t.$$

Since $1 \leq t \leq p$,

$$p^2 < p^2 + t \leq p^2 + p < p^2 + 2p + 1 = (p+1)^2.$$

So $p^2 + t$ is strictly between two consecutive squares, hence not a square. Therefore $xy^2z \notin L_2$, contradiction.

So L_2 is not regular.

$L_3 = \{a^p \mid p \text{ prime}\}$ **is not regular.** Assume L_3 is regular with pumping length p . Pick a prime number $q > p$ and let

$$s = a^q \in L_3.$$

For any split $s = xyz$ with $|xy| \leq p$, $|y| \geq 1$, write $y = a^t$ with $1 \leq t \leq p$.

Choose pumping index $i = q+1$. Then

$$|xy^{q+1}z| = q + (q)t = q(1+t).$$

Because $q \geq 2$ and $1+t \geq 2$, this length is composite, not prime. So $xy^{q+1}z \notin L_3$, contradiction.

So L_3 is not regular.

$L_4 = \{a^k b^m a^{k+m} \mid k, m \geq 0\}$ **is not regular.** Assume L_4 is regular with pumping length p . Choose

$$s = a^p b^p a^{2p} \in L_4$$

by taking $k = p$, $m = p$.

For any split $s = xyz$ with $|xy| \leq p$ and $|y| \geq 1$, y lies in the first block of a 's, so $y = a^t$ with $t \geq 1$.

Pump down with $i = 0$:

$$xz = a^{p-t} b^p a^{2p}.$$

In this new string, the first block has length $k' = p-t$ and middle block has length $m' = p$. If it were in L_4 , the last block would have to be

$$k' + m' = (p-t) + p = 2p-t,$$

but the actual last block length is $2p$. Not equal. So $xz \notin L_4$, contradiction.

So L_4 is not regular.

3.5.2 Discord Question

For pumping lemma proofs, how do you usually choose the string so the contradiction is easiest to show?

3.6 Week 6

3.6.1 Homework 6: Turing Machines on Binary Strings

We use the language

$$L = \{10^n \mid n \in \mathbb{N}\}.$$

That means: a valid string starts with one 1, and after that has only 0's.

Part 1. Accept L and output 10^{n+1} for input 10^n . Simple TM idea:

1. Start at the left end. If first symbol is not 1, reject.
2. Move right. While reading 0, keep moving right.
3. If you see another 1 after the first symbol, reject.
4. When you reach blank, write one 0 and halt-accept.

Why this works: if input is 10^n , the head reaches the end and adds one more 0, so result is 10^{n+1} .

Part 2. Accept L and output 1 for input 10^n . Simple TM idea:

1. First, do the same check as Part 1 (must be one 1 followed only by 0's). If not, reject.
2. Move head back to the first cell.
3. Keep the first 1.
4. Move right and change every 0 to blank until end of string.
5. Halt-accept.

Why this works: all zeros are erased, so only 1 remains.

Part 3. Accept all binary strings and swap every 0 and 1. Language here is $\{0,1\}^*$ (all binary strings).

Simple TM idea:

1. Start at left end.
2. If symbol is 0, write 1 and move right.
3. If symbol is 1, write 0 and move right.
4. If symbol is blank, halt-accept.

Why this works: each bit is flipped exactly once, so the output is the input with 0/1 swapped everywhere.

3.6.2 Homework 6: Turing Machines on Binary Strings (Exercise B)

Part 1. Unary addition on input 10^n10^m , output $10^n10^m10^{n+m}$. Language:

$$L = \{10^n10^m \mid n, m \in \mathbb{N}\}.$$

Simple TM idea:

1. First, verify the format is one 1, then zeros, then one 1, then zeros. If not, reject.
2. Move to the first 0 block. For each unmarked 0, mark it as X , go to the end, and write a new 0.
3. Return and do the same for the second 0 block (mark each used 0 as Y).
4. Restore $X \rightarrow 0$ and $Y \rightarrow 0$, then halt-accept.

Why this works: one new 0 is appended for every 0 in both blocks, so the last block has length $n + m$.

Part 2. Double any binary string: input w , output ww . Language accepted: $\{0, 1\}^*$.

Simple TM idea:

1. Scan from left to right for an unmarked symbol.
2. If it is 0, mark it (say X), go to end, write 0; if it is 1, mark it (say Y), go to end, write 1.
3. Return to the left and repeat until all original symbols are marked.
4. Restore marks $X \rightarrow 0$, $Y \rightarrow 1$, then halt-accept.

Why this works: the machine copies each original symbol to the end in the same order, so final tape is exactly ww .

3.6.3 Homework 6: Turing Machines on Binary Strings (Exercise C)

Transition diagrams (simple/high-level) for the previous parts:

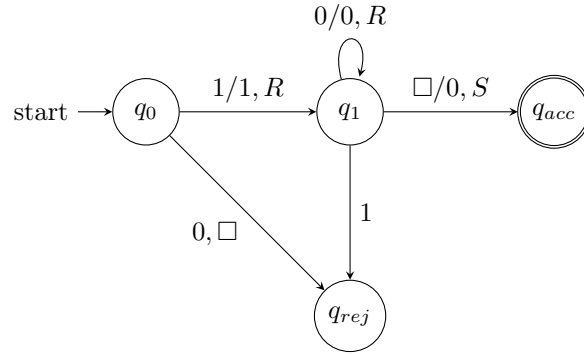


Figure 24: Exercise A1: accept 10^n and append one 0

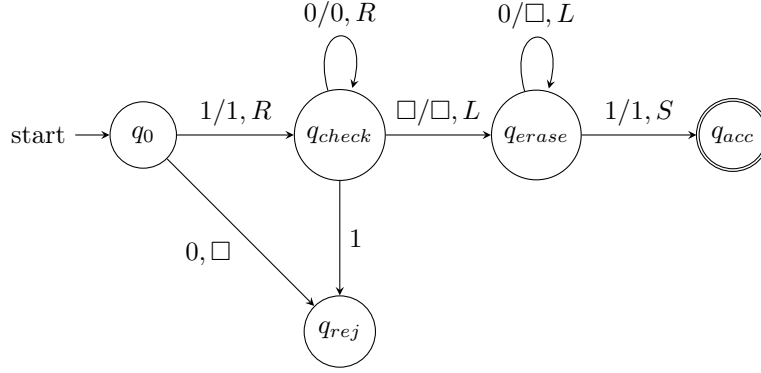


Figure 25: Exercise A2: accept 10^n and erase zeros to output 1

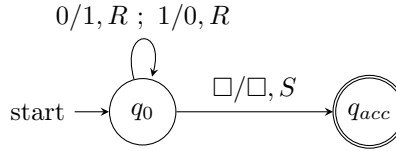


Figure 26: Exercise A3: swap every 0 and 1

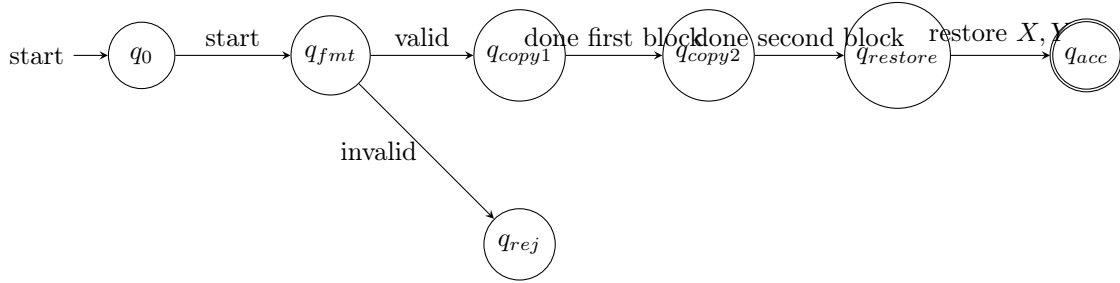


Figure 27: Exercise B1: unary addition $10^n 10^m \mapsto 10^n 10^m 10^{n+m}$ (high-level flow)

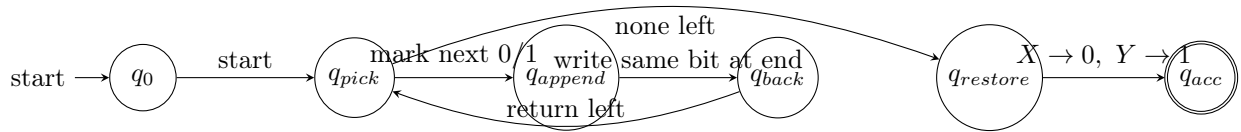


Figure 28: Exercise B2: doubling binary string $w \mapsto ww$ (high-level flow)

3.6.4 Homework 6: Turing Machines on Binary Strings (Exercise D)

Given prompt: accept binary strings that contain at least one 1, and do not halt.

Short note: With the standard TM definition, an *accept* state is halting. So "accept and never halt" is not possible in the strict sense.

The usual intended version is:

- If input has at least one 1, go to an accept state.
- If input has no 1 (only 0's or empty), loop forever.

Simple transition diagram (recognizer version):

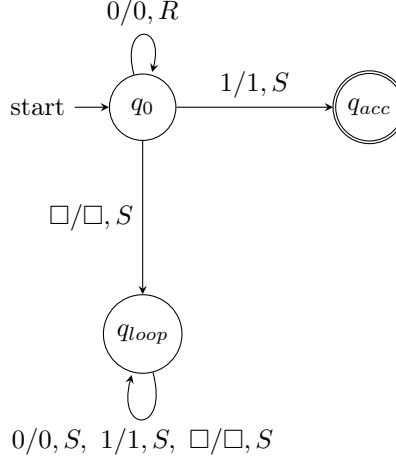


Figure 29: Exercise D: accepts strings with at least one 1, loops otherwise

3.6.5 Homework 6: Turing Machines on Binary Strings (Exercise E)

Part 1. Multitape TM: input $w \in \{0, 1\}^*$, **output** $0^n 1^m$. Here, n is the number of 0's in w , and m is the number of 1's.

Simple 3-tape idea:

- Tape 1: input w (read only).
- Tape 2: counter/output for 0's (write one 0 each time Tape 1 reads a 0).
- Tape 3: counter/output for 1's (write one 1 each time Tape 1 reads a 1).

After Tape 1 ends, copy Tape 2 then Tape 3 to the final output tape. Result is $0^n 1^m$.

Part 2 (Challenge). Multitape TM: input $10^n 10^m$, **output** $10^n 10^m 10^{nm}$. **Simple 4-tape idea:**

- Tape 1: input $10^n 10^m$.
- Tape 2: store the first zero-block (n zeros).
- Tape 3: store the second zero-block (m zeros).
- Tape 4: build the product block.

Algorithm:

1. Validate format $10^n 10^m$ and copy the two zero-blocks to Tapes 2 and 3.
2. For each 0 on Tape 2, scan Tape 3 once and append all its 0's to Tape 4.
3. When Tape 2 is finished, Tape 4 has exactly nm zeros.
4. Output $10^n 10^m 10^{nm}$ and halt-accept.

Why this works: repeating a copy of m zeros exactly n times gives nm zeros.

3.6.6 Homework 6: Problems on Decidability (Exercise 1)

For each statement, we say if it is true or false, with a short reason.

1. **True.** If L_1 and L_2 are decidable, then $L_1 \cup L_2$ is decidable.

Run the decider for L_1 and the decider for L_2 on input w . Accept if at least one accepts; otherwise reject.

2. **True.** If L is decidable, then $\overline{L}$ is decidable.

Just run the decider for L and flip the answer (accept $\leftrightarrow$ reject).

3. **True.** If L is decidable, then L^* is decidable.

For input w of length n , check all possible ways to split w into pieces. There are finitely many splits. For each piece, run the decider for L . Accept if some split has all pieces in L ; else reject.

4. **True.** If L_1 and L_2 are r.e., then $L_1 \cup L_2$ is r.e.

Run both recognizers in dovetail (step-by-step). If either one accepts, accept.

5. **False.** If L is r.e., it does *not* imply $\overline{L}$ is r.e.

Counterexample: $HALT = \{\langle M, w \rangle \mid M \text{ halts on } w\}$ is r.e., but $\overline{HALT}$ is not r.e.

6. **True.** If L is r.e., then L^* is r.e.

A recognizer can enumerate all splits $w = x_1 \cdots x_k$ and dovetail runs of the recognizer for L on each x_i . If there is a valid split with all $x_i \in L$, it will eventually be found and accepted.

3.6.7 Discord Question

For the Turing machine problems, is there a general trick for deciding whether to mark symbols, erase them, or just scan once?

3.7 Week 7

3.7.1 Homework 7: Decidable vs r.e. vs co-r.e.

Given:

$$L_1 := \{M \mid M \text{ halts on itself}\}, \quad L_2 := \{\langle M, w \rangle \mid M \text{ halts on } w\}, \quad L_3 := \{\langle M, w, k \rangle \mid M \text{ halts on } w \text{ in at most } k \text{ steps}\}.$$

1) L_1

- **r.e.: yes.** Simulate M on input $\langle M \rangle$. If it halts, accept.
- **decidable: no.** This is a halting-type problem (same difficulty as $HALT$), so no decider exists.
- **co-r.e.: no.** If both L_1 and $\overline{L_1}$ were r.e., then L_1 would be decidable, contradiction.

2) L_2

- **r.e.: yes.** This is exactly $HALT$. Simulate M on w ; if it halts, accept.
- **decidable: no.** $HALT$ is undecidable.
- **co-r.e.: no.** $\overline{HALT}$ is not r.e.

3) L_3

- **decidable: yes.** Simulate M on w for exactly k steps. Accept if halts within those steps; otherwise reject.
- **r.e.: yes.** Every decidable language is r.e.
- **co-r.e.: yes.** Every decidable language has decidable complement, hence co-r.e.

3.7.2 Homework 7: Asymptotic Notation (Exercise 2)

Let $f, g, h : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$.

1. $f \in O(f)$.

Take $c = 1$ and any n_0 . Then $f(n) \leq c f(n)$ for all $n \geq n_0$.

2. If $c > 0$ is constant, then $O(c \cdot f) = O(f)$.

Multiplying by a positive constant only rescales values. So any upper bound of the form $C c f(n)$ is still just a constant times $f(n)$, and vice versa. Therefore the two classes are equal.

3. If $f \leq g$ for large enough inputs, then $O(f) \subseteq O(g)$.

Let $u \in O(f)$. Then $u(n) \leq C f(n)$ eventually. Also $f(n) \leq g(n)$ eventually, so $u(n) \leq C g(n)$ eventually. Hence $u \in O(g)$.

4. If $O(f) \subseteq O(g)$, then $O(f + h) \subseteq O(g + h)$.

Since $f \in O(f)$ and $O(f) \subseteq O(g)$, we get $f \in O(g)$, so for some $A > 0$ and large n :

$$f(n) \leq A g(n).$$

If $u \in O(f + h)$, then for some $C > 0$ and large n :

$$u(n) \leq C(f(n) + h(n)) \leq C(Ag(n) + h(n)) \leq C \max\{A, 1\}(g(n) + h(n)).$$

So $u \in O(g + h)$.

5. If $h(n) > 0$ for all n and $O(f) \subseteq O(g)$, then $O(f \cdot h) \subseteq O(g \cdot h)$.

As above, $f \in O(g)$, so $f(n) \leq A g(n)$ eventually. Let $u \in O(fh)$. Then $u(n) \leq C f(n) h(n) \leq C A g(n) h(n)$ eventually (valid since $h(n) > 0$). Hence $u \in O(gh)$.

3.7.3 Homework 7: Asymptotic Notation (Exercise 3)

Let $i, j, k, n \in \mathbb{N}$.

1. If $j \leq k$, then $O(n^j) \subseteq O(n^k)$.

For $n \geq 1$, $n^j \leq n^k$. Apply Exercise 2(3).

2. If $j \leq k$, then $O(n^j + n^k) \subseteq O(n^k)$.

For $n \geq 1$, $n^j + n^k \leq 2n^k$, so $n^j + n^k \in O(n^k)$. Therefore $O(n^j + n^k) \subseteq O(n^k)$.

3. Assume $a_i \geq 0$ for all i and $a_k > 0$. Then

$$O\left(\sum_{i=0}^k a_i n^i\right) = O(n^k).$$

For $n \geq 1$,

$$\sum_{i=0}^k a_i n^i \leq \left(\sum_{i=0}^k a_i\right) n^k,$$

so the polynomial is in $O(n^k)$. Also, since $a_k > 0$,

$$a_k n^k \leq \sum_{i=0}^k a_i n^i,$$

which gives the matching lower-order comparison, so the classes are equal.

4. $O(\log n) \subseteq O(n)$.

For $n \geq 2$, $\log n \leq n$, so inclusion follows.

5. $O(n \log n) \subseteq O(n^2)$.

For $n \geq 2$, $\log n \leq n$, hence $n \log n \leq n^2$.

3.7.4 Homework 7: Asymptotic Notation (Exercise 4)

Relationships between the classes:

1. $O(\sqrt{n}) \subsetneq O(n)$.

Since $\sqrt{n} \leq n$ for $n \geq 1$, inclusion holds. It is strict because $n \notin O(\sqrt{n})$.

2. $O(n^2) \subsetneq O(2^n)$.

Exponential growth dominates polynomial growth, so $n^2 \in O(2^n)$, and strictness holds since $2^n \notin O(n^2)$.

3. $O(\log n) \subsetneq O((\log n)^2)$.

For $n \geq 2$, $\log n \leq (\log n)^2$. Strictness holds because $(\log n)^2 \notin O(\log n)$.

4. $O(2^n) \subsetneq O(3^n)$.

Since $2^n \leq 3^n$, inclusion holds. Strictness holds because $3^n/2^n = (3/2)^n \rightarrow \infty$, so $3^n \notin O(2^n)$.

5. $O(\log_2 n) = O(\log_3 n)$.

By change of base,

$$\log_2 n = \frac{\log_3 n}{\log_3 2},$$

so they differ only by a positive constant factor.

3.7.5 Homework 7: Asymptotic Notation (Exercise 5)

Classic sorting comparisons:

1. Bubble sort vs Insertion sort

- Bubble sort: worst/average $\Theta(n^2)$, best $\Theta(n)$ only with early-stop optimization.
- Insertion sort: worst/average $\Theta(n^2)$, best $\Theta(n)$ on already sorted input.

So asymptotically they are the same class in worst case: both in $O(n^2)$.

2. Insertion sort vs Merge sort

- Insertion sort: $\Theta(n^2)$ worst/average.
- Merge sort: $\Theta(n \log n)$ worst/average/best.

Merge sort is asymptotically better for large n because $n \log n$ grows slower than n^2 .

3. Merge sort vs Quick sort

- Merge sort: always $\Theta(n \log n)$.
- Quick sort: average $\Theta(n \log n)$, worst $\Theta(n^2)$ (bad pivots).

So in average case they are similar, but merge sort has the stronger worst-case guarantee. (In practice, quick sort is often very fast due to low constant factors and cache behavior.)

3.7.6 Discord Question

For decidable vs. r.e. vs. co-r.e., is the main idea just checking whether you can simulate the machine forever or only for a bounded number of steps?

3.8 Week 8

3.8.1 Exercise 1

Rewrite the following formulas in CNF:

1. $\varphi_1 := \neg((a \wedge b) \vee (\neg c \wedge d))$

Using De Morgan's laws,

$$\neg((a \wedge b) \vee (\neg c \wedge d)) = \neg(a \wedge b) \wedge \neg(\neg c \wedge d) = (\neg a \vee \neg b) \wedge (c \vee \neg d).$$

So the CNF is $(\neg a \vee \neg b) \wedge (c \vee \neg d)$.

2. $\varphi_2 := \neg((p \vee q) \rightarrow (r \wedge \neg s))$

First rewrite the implication:

$$(p \vee q) \rightarrow (r \wedge \neg s) = \neg(p \vee q) \vee (r \wedge \neg s).$$

Then negate and simplify:

$$\neg(\neg(p \vee q) \vee (r \wedge \neg s)) = (p \vee q) \wedge \neg(r \wedge \neg s) = (p \vee q) \wedge (\neg r \vee s).$$

So the CNF is $(p \vee q) \wedge (\neg r \vee s)$.

3.8.2 Exercise 2

Are the following formulas satisfiable? For each one, if yes, give an assignment that makes it true; if not, explain why there does not exist any.

1. $\psi_1 := (a \vee \neg b) \wedge (\neg a \vee b) \wedge (\neg a \vee \neg b)$

Yes. One satisfying assignment is $a = \text{false}$, $b = \text{false}$. With both variables false, all three clauses are true.

2. $\psi_2 := (\neg p \vee q) \wedge (\neg q \vee r) \wedge \neg(\neg p \vee r)$

No. The last clause means $p = \text{true}$ and $r = \text{false}$. Then $(\neg p \vee q)$ forces $q = \text{true}$, but $(\neg q \vee r)$ would then require $r = \text{true}$, which is a contradiction.

3. $\psi_3 := (x \vee y) \wedge (\neg x \vee y) \wedge (x \vee \neg y) \wedge (\neg x \vee \neg y)$

No. The first two clauses force $y = \text{true}$, while the last two clauses force $y = \text{false}$. So no assignment can satisfy all four clauses.

3.8.3 Discord Question

When turning a formula into CNF, is the main strategy just pushing negations inward and then distributing OR over AND?

3.9 Week 9

3.9.1 Homework 9: General vs. Horn Resolution

1a) General resolution

Given

$$\varphi := (a \vee b) \wedge (\neg a \vee c) \wedge (\neg b \vee c) \wedge (\neg c \vee d) \wedge \neg d.$$

Convert to clauses:

$$C_1 = (a \vee b), \quad C_2 = (\neg a \vee c), \quad C_3 = (\neg b \vee c), \quad C_4 = (\neg c \vee d), \quad C_5 = (\neg d).$$

Resolution steps:

$$\begin{aligned} C_6 &: \text{Res}(C_4, C_5) = (\neg c), \\ C_7 &: \text{Res}(C_2, C_6) = (\neg a), \\ C_8 &: \text{Res}(C_3, C_6) = (\neg b), \\ C_9 &: \text{Res}(C_1, C_7) = (b), \\ C_{10} &: \text{Res}(C_9, C_8) = \square. \end{aligned}$$

Since we derived the empty clause $\square$, the formula φ is **unsatisfiable**.

1b) Unit resolution on a Horn formula

Given

$$\psi = p \wedge q \wedge (\neg p \vee \neg q \vee r) \wedge (\neg r \vee s) \wedge \neg s.$$

Clause form:

$$D_1 = (p), \quad D_2 = (q), \quad D_3 = (\neg p \vee \neg q \vee r), \quad D_4 = (\neg r \vee s), \quad D_5 = (\neg s).$$

This is a Horn formula because each clause has *at most one positive literal*:

- D_1 has one positive literal (p) ,
- D_2 has one positive literal (q) ,
- D_3 has one positive literal (r) ,
- D_4 has one positive literal (s) ,
- D_5 has no positive literal.

Unit-resolution steps (each step uses at least one unit clause):

$$\begin{aligned} D_6 &: \text{Res}(D_4, D_5) = (\neg r), \\ D_7 &: \text{Res}(D_3, D_1) = (\neg q \vee r), \\ D_8 &: \text{Res}(D_7, D_2) = (r), \\ D_9 &: \text{Res}(D_8, D_6) = \square. \end{aligned}$$

So ψ is also **unsatisfiable**.

2 Exercise 2: Drone control system

Let the propositional variables be:

G : GPS lock is valid, B : battery is healthy, M : manual override key is present, E : motor power is enabled.

Policy formalization

$$(P1) (G \wedge B) \rightarrow E,$$

$$(P2) M \rightarrow E,$$

$$(P3) \neg B \rightarrow \neg E.$$

Incident log formalization

$$(L1) G, \quad (L2) \neg B, \quad (L3) M, \quad (L4) E.$$

Clause form (CNF)

$$C_1 = (\neg G \vee \neg B \vee E) \quad \text{from (P1),}$$

$$C_2 = (\neg M \vee E) \quad \text{from (P2),}$$

$$C_3 = (B \vee \neg E) \quad \text{from (P3),}$$

$$C_4 = (G), \quad C_5 = (\neg B), \quad C_6 = (M), \quad C_7 = (E) \quad \text{from log.}$$

Resolution check for consistency

Use $C_3 = (B \vee \neg E)$ and $C_5 = (\neg B)$:

$$\text{Res}(C_3, C_5) = (\neg E).$$

Then resolve with $C_7 = (E)$:

$$\text{Res}((\neg E), C_7) = \square.$$

Since we derive the empty clause, the policy together with the incident log is **inconsistent** (unsatisfiable).

Where is the conflict?

Policy rule (P3) says: if battery is unhealthy ($\neg B$), motor must be blocked ($\neg E$). But the log states both $\neg B$ and E . So the direct conflict is:

$$\neg B \rightarrow \neg E \quad \text{vs.} \quad (\neg B \wedge E) \quad \text{in the log.}$$

The manual override (M) supports E via (P2), but it does not remove the contradiction with safety rule (P3).

3.9.2 Discord Question

With resolution, how do you tell when you're close to getting the empty clause instead of just going in circles with more clauses?

3.10 Week 10

3.10.1 Homework 10: Exercises 4.3-1–3: Simple recurrences

4.3-1 Show that the solution of $T(n) = T(n-1) + n$ is $O(n^2)$.

Solution. Unroll the recurrence:

$$T(n) = T(n-1) + n = T(n-2) + (n-1) + n = \cdots = T(1) + \sum_{k=2}^n k.$$

The sum is a quadratic in n : $\sum_{k=1}^n k = n(n+1)/2 = \Theta(n^2)$, so

$$T(n) = T(1) + \Theta(n^2) = \Theta(n^2),$$

hence in particular $T(n) = O(n^2)$. A short induction proof can make this rigorous: assume $T(m) \leq Cm^2$ for all $m < n$ and choose C large enough to cover base cases; then $T(n) = T(n-1) + n \leq C(n-1)^2 + n \leq Cn^2$ for an appropriate C .

4.3-2 Show that the solution of $T(n) = T(\lceil n/2 \rceil) + 1$ is $O(\log n)$.

Solution. Each recursive call cuts the problem size roughly in half and adds a constant cost 1. The recursion depth is the number of times we can halve n until it reaches 1, which is $\lceil \log_2 n \rceil$. Adding the constant work at each level gives

$$T(n) = \Theta(\log n),$$

so $T(n) = O(\log n)$. Formally, by induction assume $T(m) \leq C \log m$ for all $m < n$; then

$$T(n) = T(\lceil n/2 \rceil) + 1 \leq C \log(\lceil n/2 \rceil) + 1 \leq C \log n$$

for a suitable constant C and all $n \geq 2$.

4.3-3 We saw that the solution of $T(n) = 2T(\lfloor n/2 \rfloor) + n$ is $O(n \log n)$. Show that the solution of this recurrence is also $\Omega(n \log n)$. Conclude that the solution is $\Theta(n \log n)$.

Solution. Apply the Master Theorem with $a = 2$, $b = 2$, and $f(n) = n$. Then $n^{\log_b a} = n^{\log_2 2} = n$, so $f(n) = \Theta(n^{\log_b a})$. By the Master Theorem (case 2) the solution is

$$T(n) = \Theta(n \log n).$$

An elementary recursion-tree argument gives the same result: at level i there are 2^i subproblems of size about $n/2^i$, each contributing $n/2^i$, so the cost at level i is $2^i \cdot (n/2^i) = n$. There are about $\log_2 n$ levels, so total cost is $n \cdot \log_2 n = \Theta(n \log n)$. To show the lower bound $\Omega(n \log n)$, note every level contributes at least some constant times n , and there are $\Theta(\log n)$ levels, giving the lower bound.

3.10.2 Discord Question

For recurrence relations, is unrolling usually the easiest way to tell whether it's going to be like n^2 or $n \log n$?

3.11 Week 11

3.11.1 Homework 11: Exercises 4.4-2 and 4.4-5: Recursion Tree and Substitution Method

4.4-2 Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = T(n/2) + n^2$. Use the substitution method to verify your answer.

Solution.

Recursion Tree: Each level i costs $n^2/4^i$. Sum: $n^2(1 + 1/4 + 1/16 + \dots) = \frac{4n^2}{3} = O(n^2)$.

Substitution: Guess $T(n) \leq cn^2$. With $c = 4$: $T(n) = T(n/2) + n^2 \leq \frac{n^2}{4} + n^2 \leq 4n^2$ ✓

4.4-5 Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = T(n-1) + T(n/2) + n$. Use the substitution method to verify your answer.

Solution.

Recursion Tree: The $T(n-1)$ branch dominates (depth n). Each level costs $O(n)$, so total is $O(n^2)$.

Substitution: Guess $T(n) \leq cn^2$ with $c = 4$. Then $c(n-1)^2 + c(n/2)^2 + n = \frac{5cn^2}{4} - 2cn + c + n \leq 4n^2$ for all $n \geq 1$ ✓

3.11.2 Exercise 4.5-1: Master Method for Multiple Recurrences

4.5-1 Use the master method to give tight asymptotic bounds for the following recurrences.

a. $T(n) = 2T(n/4) + 1$

Solution: Apply Master Theorem with $a = 2$, $b = 4$, $f(n) = 1$.

$$n^{\log_b a} = n^{\log_4 2} = n^{1/2} = \sqrt{n}.$$

Since $f(n) = 1 = o(\sqrt{n})$ (case 1), we have $T(n) = \Theta(\sqrt{n})$.

b. $T(n) = 2T(n/4) + \sqrt{n}$

Solution: $a = 2$, $b = 4$, $f(n) = \sqrt{n}$, so $n^{\log_b a} = \sqrt{n}$. Since $f(n) = \Theta(\sqrt{n})$ (case 2 with $k = 0$), we have $T(n) = \Theta(\sqrt{n} \log n)$.

c. $T(n) = 2T(n/4) + n$

Solution: $a = 2$, $b = 4$, $f(n) = n$, so $n^{\log_b a} = \sqrt{n}$. Since $f(n) = n = \omega(\sqrt{n})$ and $2 \cdot (n/4) = n/2 \leq cn$ for $c < 1$ (case 3), we have $T(n) = \Theta(n)$.

d. $T(n) = 2T(n/4) + n^2$

Solution: $a = 2$, $b = 4$, $f(n) = n^2$, so $n^{\log_b a} = \sqrt{n}$. Since $f(n) = n^2 = \omega(\sqrt{n})$ and $2 \cdot (n/4)^2 = n^2/8 \leq cn^2$ for $c < 1$ (case 3), we have $T(n) = \Theta(n^2)$.

3.11.3 Exercise 4.5-4: Master Method Applicability

4.5-4 Can the master method be applied to the recurrence $T(n) = 4T(n/2) + n^2 \ln n$? Why or why not? Give an asymptotic upper bound for this recurrence.

Solution.

Master Method Applicability:

The Master Theorem requires $f(n)$ to be a polynomial. Here $f(n) = n^2 \ln n$ has a logarithmic factor. Since $n^{\log_b a} = n^2$, the function $n^2 \ln n$ is not polynomially larger or smaller than n^2 , so no case applies. The Master Theorem **cannot be applied**.

Upper Bound:

Use the recursion tree. Cost at level i is $4^i(n/2^i)^2 \ln(n/2^i) = n^2(\ln n - i \ln 2)$. With depth $\log_2 n$:

$$T(n) = \sum_{i=0}^{\log_2 n} n^2(\ln n - i \ln 2) = \Theta(n^2(\ln n)^2).$$

3.11.4 Discord Question

In recursion trees, how do you decide whether the top level or the total of all the levels is the part that matters most?

3.12 Week 12

3.12.1 Homework 12: Indicator Random Variables

5.2-3 Use indicator random variables to compute the expected value of the sum of n dice.

Solution.

Let X_i be the value on die i . The total is $S = X_1 + X_2 + \cdots + X_n$.

Using linearity of expectation:

$$E[S] = E[X_1] + E[X_2] + \cdots + E[X_n].$$

For one die:

$$E[X_i] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5$$

So:

$$E[S] = n \cdot 3.5 = \frac{7n}{2}$$

5.2-4 Use indicator random variables to solve the hat-check problem. Each of n customers gives a hat to a hat-check person at a restaurant. The hat-check person gives the hats back to the customers in a random order. What is the expected number of customers who get back their own hat?

Solution.

Let $X_i = 1$ if customer i gets their own hat back, and $X_i = 0$ otherwise.

The total is:

$$X = X_1 + X_2 + \cdots + X_n$$

By linearity of expectation:

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_n]$$

The probability that customer i gets their own hat is $P(X_i = 1) = \frac{1}{n}$ (since there are n hats and each customer has a $1/n$ chance of getting theirs).

So:

$$E[X_i] = \frac{1}{n}$$

Therefore:

$$E[X] = n \cdot \frac{1}{n} = 1$$

The expected number of customers who get their own hat back is **1**.

3.12.2 Discord Question

For indicator random variables, is the main trick just making one variable per event and then adding up all the expectations?

3.13 Week 13

3.13.1 Homework 13: Encoding Sudoku

Exercise (Encoding Sudoku). Write up boolean formulas encoding the rules of Sudoku. For each row $r \in \{1, \dots, 9\}$, column $c \in \{1, \dots, 9\}$, and value $v \in \{1, \dots, 9\}$ we introduce a variable $x_{r,c,v}$. The variable $x_{r,c,v}$ is true if and only if the entry (r, c) has value v . Formalize each of the conditions C_k as a boolean formula.

Solution.

For brevity, we use the big quantifier notation where $\bigwedge_{i=1}^n A_i = A_1 \wedge A_2 \wedge \dots \wedge A_n$ and $\bigvee_{i=1}^n A_i = A_1 \vee A_2 \vee \dots \vee A_n$.

C_1 : Each entry has at least one value

$$C_1 := \bigwedge_{r=1}^9 \bigwedge_{c=1}^9 \bigvee_{v=1}^9 x_{r,c,v}$$

For each cell, at least one of the nine values must be true.

C_2 : Each entry has at most one value

$$C_2 := \bigwedge_{r=1}^9 \bigwedge_{c=1}^9 \bigwedge_{v=1}^8 \bigwedge_{v'=v+1}^9 (\neg x_{r,c,v} \vee \neg x_{r,c,v'})$$

For each cell, no two different values can both be true.

C_3 : Each row has all numbers

$$C_3 := \bigwedge_{r=1}^9 \bigwedge_{v=1}^9 \bigvee_{c=1}^9 x_{r,c,v}$$

For each row and each value, that value must appear in at least one column.

C_4 : Each column has all numbers

$$C_4 := \bigwedge_{c=1}^9 \bigwedge_{v=1}^9 \bigvee_{r=1}^9 x_{r,c,v}$$

For each column and each value, that value must appear in at least one row.

C_5 : Each (3×3) -block has all numbers

$$C_5 := \bigwedge_{b=1}^9 \bigwedge_{v=1}^9 \bigvee_{(r,c) \in B_b} x_{r,c,v}$$

where B_b is the set of 9 cells in block b . For each block and each value, that value must appear in at least one cell of the block.

C_6 : *The solution respects the given clues*

$$C_6 := \bigwedge_{(r,c,v) \in \text{clues}} x_{r,c,v}$$

For each given clue cell at position (r, c) with value v , we set $x_{r,c,v}$ to true.

3.13.2 Probabilistic Algorithm Analysis

Randomized Algorithm: CountLargeRolls(n)

Algorithm:

```

c <- 0
for i <- 1 to n do
  for j <- 1 to 2i do
    r <- a uniformly random integer from {1, 2, 3, 4, 5, 6}
    if r >= 5 then
      c <- c + 1
return c

```

Question 1: Big-O worst-case running time

The outer loop runs n times. For iteration i , the inner loop runs $2i$ times. Total iterations:

$$\sum_{i=1}^n 2i = 2 \cdot \frac{n(n+1)}{2} = n(n+1) = O(n^2).$$

Each iteration does constant work, so the worst-case is $\mathbf{O(n^2)}$.

Question 2: Expected value of returned value using indicator variables

Let $X_{i,j} = 1$ if the j -th roll in iteration i is ≥ 5 , and 0 otherwise.

The total count is:

$$X = \sum_{i=1}^n \sum_{j=1}^{2i} X_{i,j}$$

By linearity of expectation:

$$E[X] = \sum_{i=1}^n \sum_{j=1}^{2i} E[X_{i,j}]$$

For each roll, $P(\text{roll} \geq 5) = P(\text{roll} = 5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$.

So $E[X_{i,j}] = \frac{1}{3}$ for all i, j . Therefore:

$$E[X] = \sum_{i=1}^n \sum_{j=1}^{2i} \frac{1}{3} = \frac{1}{3} \sum_{i=1}^n 2i = \frac{1}{3} \cdot n(n+1) = \frac{\mathbf{n(n+1)}}{\mathbf{3}}$$

Question 3 (Bonus): What's wrong with the student's reasoning?

The student says the expected value is $n \cdot \frac{2}{6}$.

The problem is the student only counted the outer loop iterations (n) but ignored the inner loop. The inner loop in iteration i runs $2i$ times, not just once. The total number of rolls is $n(n+1)$, not n . So the correct expected value is $n(n+1) \cdot \frac{1}{3}$, not $n \cdot \frac{1}{3}$.

Question 4 (Bonus): If condition changed to $r = 6$

If we only count when $r = 6$, then $P(\text{roll} = 6) = \frac{1}{6}$.

So $E[X_{i,j}] = \frac{1}{6}$ and:

$$E[X] = \sum_{i=1}^n 2i \cdot \frac{1}{6} = \frac{1}{6} \cdot n(n+1) = \frac{\mathbf{n(n+1)}}{\mathbf{6}}$$

The expected value would be half of the previous answer since the probability of rolling a 6 is half the probability of rolling 5 or 6.

3.13.3 Discord Question

For that random-roll algorithm, why isn't the expected value just the probability of rolling 5 or 6 times the number of outer-loop iterations?

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.